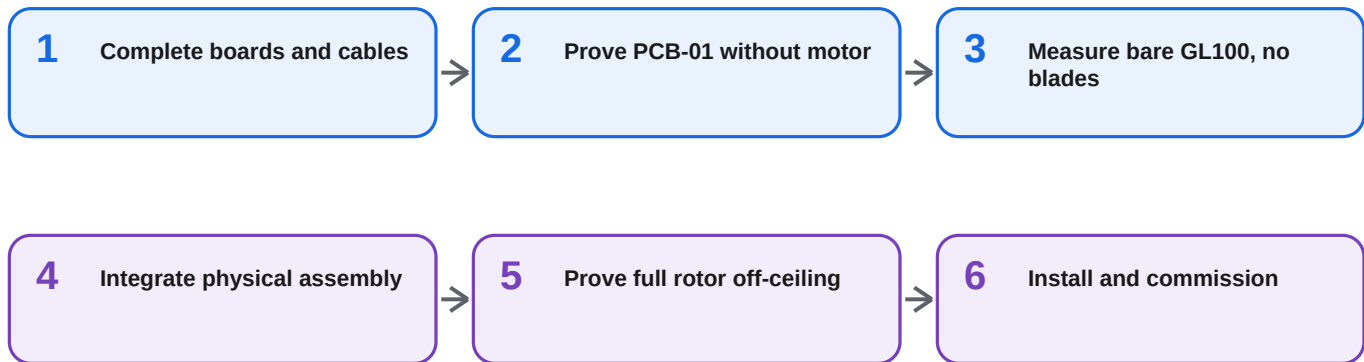


HOW TO USE THIS BINDER

Use one sheet at the bench. Check each box only after the action is complete. Record measurements in `testing/test-matrix.csv`. A HOLD badge means prerequisites are unresolved; do not improvise past it.

DEPENDENCY SPINE



CURRENT CHECKPOINT

- ☐ PCB-01 and PCB-02 hand-population complete and inspected.
- ☐ Two primary ceiling holes drilled and cleaned; no anchors installed.
- ☐ PCB-01 passes PCB-01 through PCB-04 and TACH-01 without the motor.

NON-NEGOTIABLE BOUNDARIES

- ☐ No blades during bare-motor characterization.
- ☐ No permanent ceiling-plate installation before off-ceiling proof, tether, bearing, and approval gates close.
- ☐ Do not drill the tether hole until location, spacing, and termination are resolved.
- ☐ Do not design ENC-100 before EB-100 and real cable bends exist.
- ☐ Never bypass the 180 RPM controller limit or 200 RPM analog trip without the written guarded two-person procedure.

1A Complete PCB-01 and PCB-02

FIELD GUIDE

TOOLS

ESD mat - microscope - temperature-controlled iron/hot air - flux - solder wick - DMM

PCB-01 HAND-POPULATION

- ☐ Inspect the assembled board for shipping damage and solder bridges.
- ☐ Fit C1 and C2: Panasonic FR 470 uF / 50 V. Observe polarity and silkscreen.
- ☐ Fit J1: Molex 43045-0200 power header.
- ☐ Fit J2: Molex 43650-0300 phase header.
- ☐ Fit U8: LM2907M/NOPB. Match pin 1 to board marking.
- ☐ Bridge C34 with KEMET C1206C104K3GACTU, 100 nF C0G, across the 0603 site.
- ☐ Bridge F1 pads. The wall-side 3 A fuse is then mandatory.
- ☐ Inspect every hand joint and verify no RAW24-to-ground short.

PCB-02 HAND-ASSEMBLY

- ☐ Fit U1: DRV5033FAQDBZR, with package orientation matching PCB marking.
- ☐ Fit C1: 100 nF / 50 V X7R 0603 local bypass.
- ☐ Fit J1: S3B-PH-K-S side-entry JST-PH.
- ☐ Keep every component on the magnet-facing side; solder before mounting to BR-100.
- ☐ Inspect under magnification and continuity-check 3V3, HALL_TACH, and AGND.

DO NOT POWER either board until 1B harness polarity and the unpowered continuity checks are complete.

STOP / HANDOFF

Both boards are fully populated, photographed, and pass unpowered short/polarity inspection.

CONNECTOR TRUTH

CONNECTOR	FIRST PINS	REMAINDER
J1 POWER	1 RAW24	2 0V
J2 MOTOR	1 W	2 V 3 U
J3 / PCB-02 J1	1 3V3	2 HALL_TACH 3 AGND
J4 TEMP	1 TEMP_SENSE	2 AGND
J5 I2C	GND, 3V3	SDA, SCL
J7 PROGRAM	3V3, TX, RX	EN, BOOT, GND

BUILD CHECKLIST

- ☐ Power harness: Belden 5300UE 18/2 to Micro-Fit; preserve +24 V and 0 V end-to-end.
- ☐ Motor harness: use 18 AWG contacts; label connector positions W, V, U even if motor flying leads are initially unlabeled.
- ☐ Hall harness: PHR-3 to PHR-3 straight-through, using molded No.1 marks at both ends.
- ☐ Temperature harness: PHR-2 to the NTC ring lug; exact motor attachment remains TBD.
- ☐ Programming lead: J7 pin 1 is board 3V3, not a power input.
- ☐ Pull-test every crimp, then continuity-check each contact and verify no cross-continuity.
- ☐ Record harness length, wire colors, and contact orientation before sleeving.

J7 pin 1 ties directly to board 3V3. Never connect a programmer that actively drives 3.3 V while 24 V is connected.

A swapped Hall contact can leave the tach silently dead. Require 1-to-1, 2-to-2, 3-to-3 and no cross-continuity before first power.

STOP / HANDOFF Every harness is labeled, pull-tested, and has a signed continuity result. No device is powered yet.

SAFE TEST SETUP

- ☐ Disconnect motor phases and keep the rotor absent.
- ☐ Use the fused 24 V bench path with current limiting and a physical cutoff within reach.
- ☐ Connect console/programmer without driving board 3V3 from J7.
- ☐ Place probes before power; use spring ground for VM transient measurements.

CHECK IN THIS ORDER

- ☐ PCB-01: power briefly; verify 24 V, 3.3 V, AVDD, DVDD, DRVOFF, current draw, and no abnormal heating.
- ☐ PCB-03: force PGOOD, WDO, manual clear, and OVERSPEED_N low one at a time. DRVOFF must rise without ESP help.
- ☐ PCB-03B: static WDI must time out in 1.44-1.76 s; WDO pulse about 200 ms. Falling edges at 2 Hz prevent timeout.
- ☐ PCB-03D: repeat cold and slow-ramp power-up 20 times; U6 /PRE must release after TACH_PGOOD_N.
- ☐ PCB-04: remove and restore 24 V. Fan remains disabled for at least 10 s and until a new command plus explicit arm.
- ☐ TACH-01: inject 0-3.3 V at HALL_TACH with bridge disabled, 1-3% duty, and at least 30 s settling.
- ☐ Record reset near 3.00 Hz and trip near 3.33 Hz; the persistent lock resets only on genuine power cycle.

PASS LIMITS

VM insertion target: ≤ 35 V; coast/cutoff/stall/reversal checks occur later with the guarded motor. Watchdog: 1.44-1.76 s. Power-up: 20/20 correct presets. Tach: reset about 3.00 Hz, trip about 3.33 Hz. Any automatic re-arm blocks release.

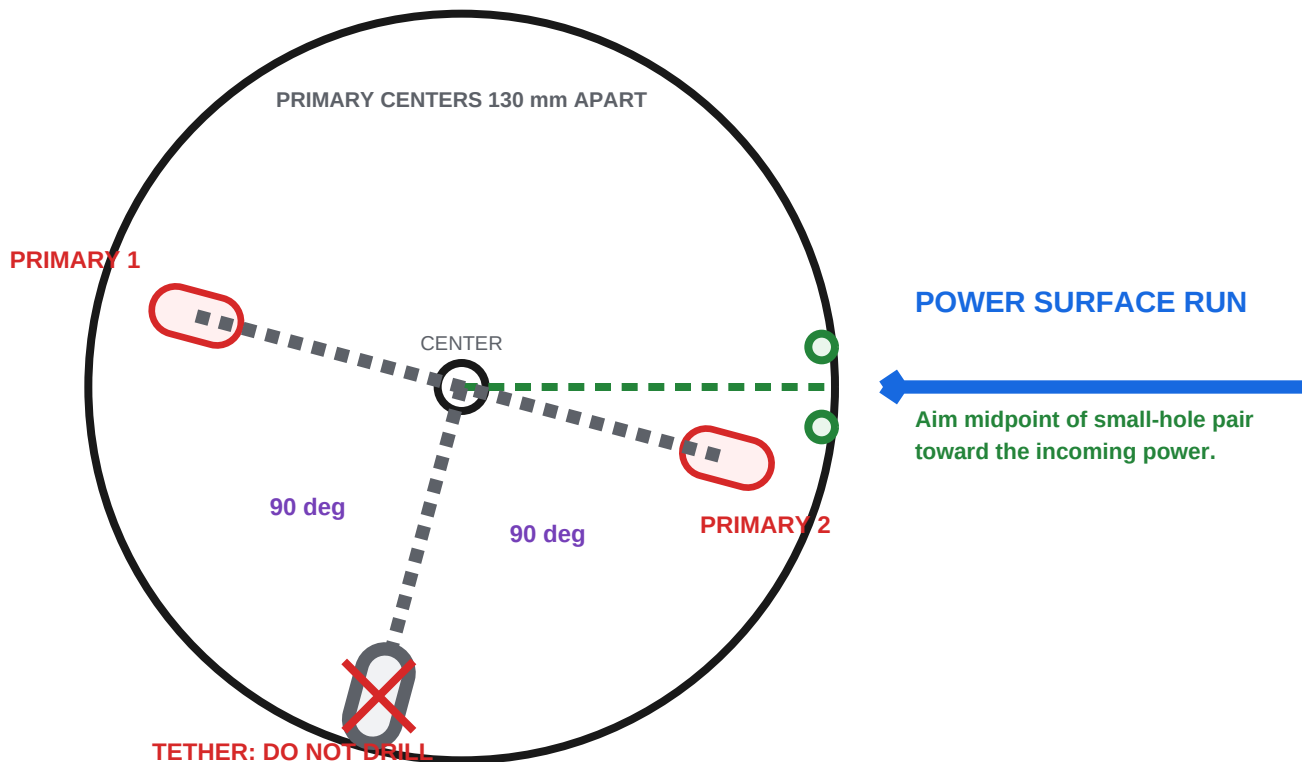
STOP / HANDOFF

PCB-01 through PCB-04 and TACH-01 have recorded pass/fail results. Do not connect the motor after a failure.

2A Ceiling Orientation and Primary Holes

FIELD GUIDE

TEMPLATE ORIENTATION - VIEWED FROM THE ROOM



DRILL AND CLEAN

- ☐ Confirm the known fan center and orient the template as shown.
- ☐ Mark only the centers of the two red 11 x 20 mm primary slots.
- ☐ Remove the template. Use a straight-shank ANSI bit in the M12 3404, or an SDS-plus ANSI bit only in an SDS-plus rotary hammer.
- ☐ Drill with a 3/8 in ANSI B212.15 carbide bit; never substitute 10 mm.
- ☐ Depth-stop both primary holes at about 75 mm.
- ☐ Peck-drill to clear dust and cool the tip; stop and rent SDS-plus if progress is poor.
- ☐ Brush and blow both holes clean per Simpson instructions.

PRIMARY HOLES

3/8 in x ~75 mm

NO ANCHORS YET

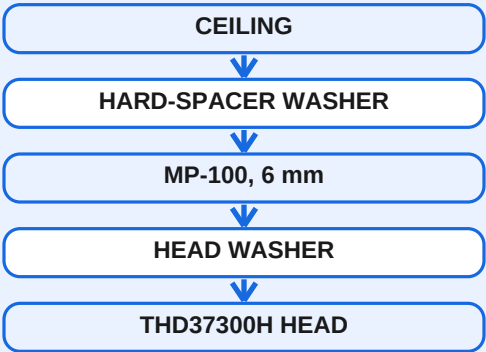
STOP / HANDOFF

Two cleaned primary holes are ready. Leave all Titen HD anchors out until permanent installation.

BENCH ASSEMBLY - CEILING FACE UP

- ☐ Insert the received SUS304 SP-100 from the ceiling face; seat flange fully in the double-D recess, metal on the pocket shoulder.
- ☐ Place three tiny spaced dots of neutral-cure RTV across the perimeter seam. Put none under the flange.
- ☐ Install 3 x ST-100 with 3 x M6 x 16 A4 flat-head screws from the ceiling face.
- ☐ Verify all three screw heads are flush or slightly below the ceiling face.
- ☐ Keep ceiling-face-up until RTV cures; tap/shake and confirm no metallic click.
- ☐ Leave off motor/carrier, rotor, catcher stack, electronics, clamps, and housing.

OVERHEAD LIFT - EACH PRIMARY STACK



Parts for both anchors

- ☐ 2 x Simpson THD37300H, 3/8 x 3 in.
- ☐ 4 x Prime-Line 9080006 washers, about 2.5 mm thick.
- ☐ 9/16 in socket and approved driver.

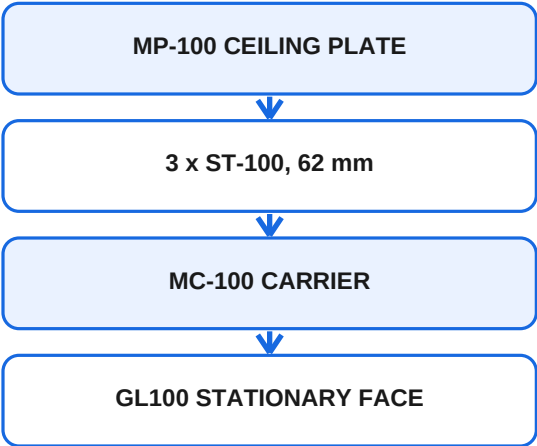
INSTALLATION CHECK

- ☐ Place one loose hard-spacer washer above MP-100 at each anchor during the lift; do not glue spacers to plate.
- ☐ Drive each anchor through head washer, MP-100 slot, hard spacer, and into cleaned concrete.
- ☐ Follow Simpson installation instructions; the screw anchor has no wedge-anchor set torque.
- ☐ Record model, spacing, embedment basis, plate seating, and any shimming under INS-01.

ONE-INSTALL INTERFACE: do not install, remove, and reinstall Titen HDs. Mounting remains blocked until off-ceiling proof, tether, motor-bearing, and installation-approval gates close.

STOP / HANDOFF MP-100 is permanently seated on hard spacers, SP-100/ST-100 are captive, and INS-01 is recorded.

STACK ORDER



Fasteners

- ☐ MP-100 to ST-100: 3 x M6 x 16 flat-head from ceiling face.
- ☐ MC-100 to ST-100: 3 x M6 x 20 A4-80 with wedge-lock pairs.
- ☐ MC-100 to GL100 rear face: 4 x M4 x 12 A4-80 on 60 mm PCD.

DRY-FIT CHECKLIST

- ☐ Confirm GL100 stationary rear face is the 60 mm M4 pattern; rotating output is the 50 mm pattern.
- ☐ Confirm the motor wire exit clears the MC-100 phase window without pinching.
- ☐ Hand-start every fastener. Stop if any screw bottoms before clamping.
- ☐ Verify the three ST-100s seat squarely and MC-100 is not rocked by finish or burrs.
- ☐ Verify M4 x 12 gives about 5.5 mm engagement and never exceeds the GL100 6.0 mm limit.
- ☐ Use the released joint torque only. Do not infer torque from a generic stainless table.
- ☐ Apply compatible removable threadlocker and witness marks only at final released assembly.
- ☐ Record stack fit, wire clearance, and any washer/length adaptation before proceeding.

Motor M4 installation torque remains a release gate. Use a calibrated driver and the released value when available.

STOP / HANDOFF The stationary stack fits without force, rocking, wire pinch, or bottoming screws. Keep it non-powered.

ROTOR ASSEMBLY

- ☐ RH-100 pilot enters the GL100 bore by hand; never press or force it.
- ☐ Install RH-100 with 4 x M4 x 10 A4 flat-head screws from the underside into the 50 mm rotating pattern.
- ☐ Verify every hub screw head is 0.1-0.2 mm subflush.
- ☐ Fit one magnet and two mass-matched brass slugs in the three r76 pockets; epoxy is the retention.
- ☐ Match each complete insert-plus-epoxy station within 0.01 g.
- ☐ Install three BP-100 v3 blades using their printed locating pins and 4 x M5 x 22 bolts per blade into the captured M5 nuts.

CATCHER STACK - FROM ROTOR DOWN

RH-100 ROTATING HUB



2.5 +/- 0.5 mm RUNNING GAP



KD-100, OD 44 / ID 13.5



M12 CASTELLATED NUT



3.2 x 32 SPLIT PIN

Nut orientation

Bearing face up against KD-100. Castellated crown downward. Rotate the nut, not the keyed spindle, to align a slot with the cross-hole.

Final checks

Trim split-pin legs. Confirm no catcher contact or witness marks after rotation.

BEFORE ANY POWER

- ☐ Hand-rotate through multiple revolutions; confirm no rub, click, or changing gap.
- ☐ Verify all critical fasteners have the released torque record and witness marks.
- ☐ Do not use cad/BP-100.step for current fit; the committed export is stale v2 geometry.
- ☐ Do not install the rotor over the bed before off-ceiling proof passes.

STOP / HANDOFF

Rotor and catcher stack turn freely by hand with correct gap, retention, and no contact.

HALL SENSOR GEOMETRY - RELEASED

- ☐ PCB-02 components face the rotor magnet; J1 and cable exit outboard.
- ☐ Sensor element and magnet centerline are both at radius 76.0 +/- 0.5 mm.
- ☐ Set 2.5 mm nominal axial gap, adjustable 1.5-4.0 mm, measured from the marked SOT-23 outer face to magnet cap.
- ☐ Use PCB-02 M2 holes on 6 mm pitch. At H1 use a bare pan head or washer <=4.5 mm OD.
- ☐ Provide cable strain relief on BR-100; PCB material does not extend beyond J1.
- ☐ Hand-turn rotor: one clean pulse per revolution with magnet; no false pulses with magnet removed.

BLOCKED FABRICATION ITEMS

- ☐ BR-100 remains owner hand-fabricated; validate around the physical motor, PCB-02, cable, and Hall gap.
- ☐ Populate PCB-01 connectors before defining EB-100 and real cable bends.
- ☐ Design EB-100 before ENC-100. Reserve 110 x 80 x 25 mm plus service clearance.
- ☐ ENC-100 cable notch must remain open at the top rim and align with the 15 deg power-entry line.
- ☐ Temperature-sensor motor attachment point remains TBD.

RELEASE CHECK

- ☐ Connector bodies and cable bends clear the rotor and standoffs.
- ☐ PCB-01 has an independent retention lanyard and independent cable clamps.
- ☐ Housing half can be removed without loading or trapping the power cable.

STOP / HANDOFF

Proceed only with Hall fit-up and pulse verification. Do not fabricate EB-100 or ENC-100 from provisional geometry.

BUILD AND FLASH

```
cd firmware && cargo build
```

```
espflash flash --port /dev/cu.usbmodem2101 --non-interactive  
app/target/riscv32imac-unknown-none-elf/debug/stillair
```

```
target/debug/stillair --port /dev/cu.usbmodem2101 state
```

BARE DEV-BOARD INPUT SIMULATION

- ☐ GPIO22 to 3V3: PGOOD good.
- ☐ GPIO21 to 3V3: nFAULT idle high.
- ☐ GPIO14 to GND: ALARM idle low.
- ☐ Without real FG edges, Starting must fault NoRotation after 15 s. This is expected.

SESSION RULES

- ☐ Use one serial reader at a time. Stop raw-log capture before espflash or the CLI opens the port.
- ☐ Use `script` for any multi-step sequence. Separate CLI invocations create separate simulator sessions.
- ☐ Use `wait <state> --for <seconds>` after asynchronous commands.
- ☐ Use `stream <hz> --for <seconds>` for CSV telemetry.
- ☐ For release, `config check` must report config=verified; ok=true with config=unverified is not a pass.
- ☐ Use `config capture` only after a complete measured configuration exists.
- ☐ Treat simulator results as harness validation, never motor validation.

WARNING: The CLI only surfaces @-prefixed protocol lines. Capture raw logs for Wi-Fi/Matter failures. Never use raw reg write on a spinning motor except under an approved stopped/instrumented procedure.

STOP / HANDOFF

Firmware flashes, one CLI session communicates reliably, and scripted sequences preserve device state.

4B Bare-Motor Measurements and Config Gate

HOLD: IMAGE UNVERIFIED

PREREQUISITES

- ☐ PCB-01 through PCB-04 and TACH-01 passed.
- ☐ GL100 is rigidly guarded with NO BLADES installed.
- ☐ VM scope probe and manual cutoff are ready before power.

BARE-MOTOR WORK ALLOWED NOW

- ☐ Measure phase R and L independently with the method and connection convention recorded.
- ☐ Manually spin the GL100 and scope line-to-line BEMF; record amplitude and phase convention.
- ☐ Record the measured values beside provisional seeds 0xB1 / 0xAE / 0xCA. Do not call the seeds released.
- ☐ Confirm 20 pole pairs and inspect console, FG, Hall, watchdog, and stop behavior without raw live register writes.
- ☐ Scope only the motor operations already approved by the guarded procedure; VM target ≤ 35 V and 40 V rejects.
- ☐ Keep blades and loose hub hardware off throughout this bare-motor stage.

MPET AND GOLDEN IMAGE - HOLD

- ☐ Do not run MPET unloaded. controls.md requires the representative final rotor because unloaded MPET can produce bad Ke/Kp/Ki.
- ☐ Resolve the representative-rotor MPET procedure, then cross-check it against independent R/L/BEMF measurements.
- ☐ Populate the required D-generation fields, including safety-critical speed, ALARM, and external-watchdog settings.
- ☐ Do not use `config apply` as a release step until EEPROM completion timing and readback are proven on hardware.
- ☐ Capture the complete image only after review. Never reuse A1/C-generation register dumps.
- ☐ Require `config check` to report config=verified; ok=true with config=unverified is not a release pass.

CURRENT RECORD

R: _____ L: _____ BEMF: _____ convention: _____

Config verdict: unverified / verified image commit: _____

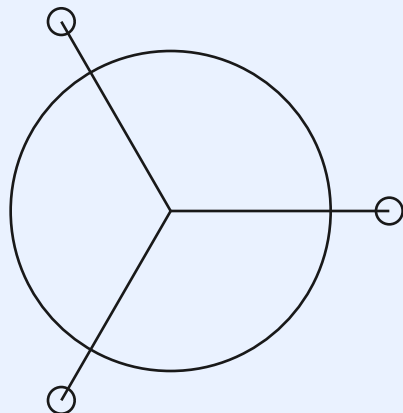
STOP / HANDOFF

Independent bare-motor measurements are recorded. MPET and the golden image remain HOLD until the loaded procedure is released.

MEASURE BEFORE POWER

- ☐ Use a released level/indexed fixture, a dial indicator readable to 0.01 mm, and a calibrated scale readable to 0.01 g.
- ☐ Define the vibration measurement and rejection criterion before any powered motion.
- ☐ Record each blade mass and first moment; target first moments within 0.5%.
- ☐ Measure RH-100 OD runout; limit ≤ 0.10 mm TIR.
- ☐ Measure all blade tips in one indexed setup; spread ≤ 0.5 mm.
- ☐ Hand-rotate and verify no catcher or housing contact.
- ☐ Correct only with the documented balance-slug method; re-measure after every change.
- ☐ Photograph witness marks and record final correction masses/locations.

VISUAL RECORD



Hub TIR: _____ mm

Tip spread: _____ mm

First-moment spread: _____ %

Correction: _____

PASS CONDITION

Hub ≤ 0.10 mm TIR, tips within 0.5 mm, first moments within 0.5%, no objectionable vibration, and no catcher contact.

STOP / HANDOFF MEC-05 passes with measurements recorded. Do not advance a rotor that only 'looks balanced.'

5B Proof Fixture and Procedure Gate

HOLD: PROCEDURE MISSING

ENGINEER BEFORE TESTING

- ☐ Release a fixture drawing with rated containment, external drive, remote cutoff/interlock, and instrument mounting.
- ☐ Write the two-person procedure: roles, callouts, abort criteria, maximum duration, safe positions, and restoration checks.
- ☐ Define the exact bypass hardware/state if a temporary electronic bypass is unavoidable.
- ☐ Define final retained mass, drop geometry, catcher/tether release mechanism, and calibrated load instrumentation.
- ☐ Define the conservative trial imbalance mass and a measurable vibration/abort criterion.
- ☐ Approve the procedure before any high-energy run or dynamic catch.

REQUIREMENTS THE PROCEDURE MUST SATISFY

- ☐ MEC-04 catcher: 1.25 kN static proof plus guarded dynamic release of the final rotor.
- ☐ MEC-03 proof requirement: external guarded drive, 216 RPM, 2 minutes each direction.
- ☐ MEC-06 imbalance: no loosening, plate motion, housing/catcher contact; no automatic shutdown claim.
- ☐ After any bypass, restore and independently reverify the 180 RPM MCF limit and 200 RPM analog trip before fixture removal.
- ☐ Never dynamically test the 270 RPM credible bypass load. It is calculation-only.
- ☐ Reject damage, opening, permanent set, deformation, balance shift, contact, or loosened witness marks.

TETHER PATH - SEPARATE HOLD

Resolve the ceiling location, ≥ 190 mm primary spacing basis, termination, exact cable/fittings, MC-100 attachment, slack, and calculated catch peak x2 requirement. Reconcile that calculated requirement with the current ≥ 4.5 kN matrix wording before INS-02 sign-off.

Do not use the MP-100 tether clearance as a drill location. Do not perform these tests on the ceiling.

STOP / HANDOFF

No test is authorized by this sheet. Release the fixture and procedures first; then issue an execution revision.

5C Full-Rotor Drive, Acoustic, and Thermal

HOLD: METHODS TO DEFINE

START AND SPEED QUALIFICATION

- ☐ With the representative final rotor guarded, run the released MPET procedure; cross-check R/L/BEMF, then complete and read back the golden image.
- ☐ Before starts, require config check to report config=verified and independently verify the 180 RPM MCF limit and 200 RPM analog trip.
- ☐ Complete 100 randomized-rest starts per direction at 30, 35, and 40 RPM.
- ☐ Repeat the start matrix at 23.3, 24.0, and 24.7 V.
- ☐ Reject any retry, reverse kick, stall, hunting, or objectionable tonal sequence.
- ☐ Verify steady operation at 30, 40, 55, 70, 120, and 170 RPM.
- ☐ Verify MCF active-control ceiling never exceeds 180 RPM.
- ☐ Test windmilling restart, reversal after verified stop, cutoff at speed, and enabled MCF faults.

ACOUSTIC AND THERMAL

- ☐ Before testing, record fixed microphone/listening position, ambient condition, sensor locations, and logging cadence.
- ☐ Listen motor-only and complete fan at 30, 40, 60, 80, 120, and 170 RPM from the fixed bed position.
- ☐ Release a sleep speed only when no identifiable motor, controller, bearing, or structural tone remains.
- ☐ Run the complete fan for 8 hours at 170 RPM on the guarded fixture.
- ☐ Record RMS phase current, MCF q-axis estimate, PWM peak, independent motor/PCB temperatures, ambient, and supply behavior.
- ☐ Normal RMS phase current <0.8 A; investigate at 1.0 A; 1.5 A limiter must not clip continuously.
- ☐ Motor <70 C, PCB <85 C, no supply dropout or overtemperature, input power <50 W.

RELEASE SUMMARY

Released minimum RPM: _____ Released sleep RPM: _____ Max qualified RPM: _____

Open anomalies / restrictions: _____

WARNING: TEMP_SENSE firmware is not implemented. Do not treat the wired NTC as active overtemperature protection.

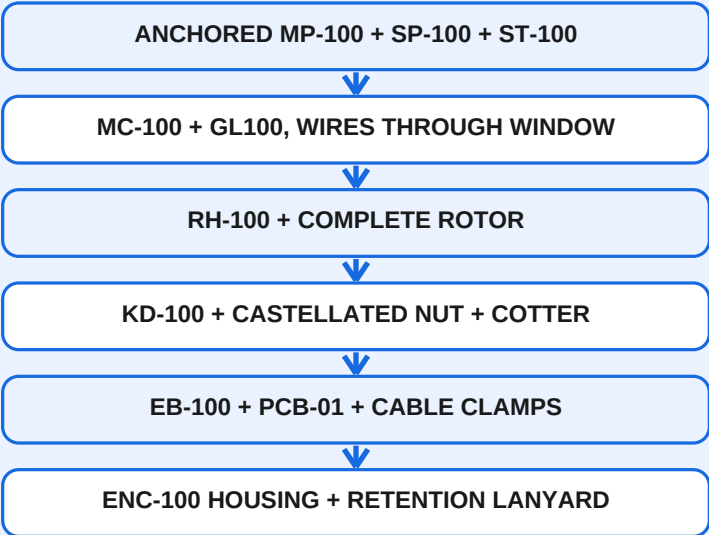
STOP / HANDOFF

All full-rotor bench tests pass and released speed limits are recorded before permanent installation.

PREREQUISITE GATE

- ☐ 2B plate installation complete and INS-01 recorded.
- ☐ 5A-5C off-ceiling release complete; tether path resolved and separately verified.
- ☐ Motor axial-bearing gate and local installation approval/certification path are closed.
- ☐ No unresolved torque, Hall-gap, EB-100, ENC-100, or cable-routing hold remains.

ASSEMBLY ORDER FROM BELOW



Checks at every layer

- ☐ No wire pinch or sharp-edge contact.
- ☐ Correct fastener, locking method, torque record, and witness mark.
- ☐ Hand rotation remains free after each rotating layer.
- ☐ Catcher gap remains 2.5 +/- 0.5 mm.
- ☐ Hall gap remains within 1.5-4.0 mm.

FINAL PHYSICAL CHECK

- ☐ Tether has 15-20 mm slack and cannot foul rotor, catcher, housing, or cable path.
- ☐ Power cable is clamped to MP-100, not the removable housing; notch remains open at top rim.
- ☐ PCB and removable housing half each have their required independent retention.
- ☐ All serviceable assemblies can be removed from below without touching the anchors.

STOP / HANDOFF

Complete assembly is secure, freely rotating by hand, correctly tethered, and ready only for limited-speed commissioning.

6B Installed Commissioning and Sign-Off

HOLD: LADDER TBD

BEFORE FIRST INSTALLED ROTATION

- ☐ Clear the room and establish a manual cutoff observer.
- ☐ Confirm plate seating, anchor washers, tether, catcher, Hall gap, wall gap, and housing clearance.
- ☐ Confirm firmware golden-image verdict is verified and power restoration remains off.
- ☐ Approve and write the speed ladder, dwell time, maximum, measurements, and abort thresholds before starting:

- ☐ Start at the released minimum speed; do not improvise a higher installed test point.

INCREASING LIMITED-SPEED CHECKS

- ☐ At each step inspect plate movement, tether behavior, catcher clearance, Hall gap, wall gap, and new resonance.
- ☐ Verify commanded speed against independent tach and recorded telemetry.
- ☐ Test normal stop and confirm coast behavior.
- ☐ Test direction reversal only through ramp-to-zero and verified stop.
- ☐ Interrupt and restore power; confirm it returns off and requires a fresh command plus explicit arm.
- ☐ Verify network loss preserves the last local speed while ESP/watchdog failure revokes drive.
- ☐ Exercise Apple Home 1%, 50%, and 100% mapping only within the installed release ceiling.

INSTALLATION RECORD

Date: _____ Installer: _____

Golden image revision / commit: _____

Installed speed ceiling: _____ RPM Released sleep speed: _____ RPM

INS-03 result: PASS / FAIL Remaining restrictions: _____

STOP / HANDOFF

INS-03 passes with no movement, contact, resonance, or clearance loss. Keep every recorded restriction in force.